Chapter 5

Sequences and Series

Only One Option Correct Type Questions

1. If the sum of first n terms of an A.P. is cn^2, then the sum of squares of these n terms is

[IIT-JEE-2009 (Paper-2)]

(A) $\dfrac{n(4n^2-1)c^2}{6}$

(B) $\dfrac{n(4n^2+1)c^2}{3}$

(C) $\dfrac{n(4n^2-1)c^2}{3}$

(D) $\dfrac{n(4n^2+1)c^2}{6}$

2. Let $a_1, a_2, a_3, ...$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is **[IIT-JEE-2012 (Paper-2)]**

(A) 22

(B) 23

(C) 24

(D) 25

3. Let $b_i > 1$ for $i = 1, 2, ..., 101$. Suppose $\log_e b_1, \log_e b_2, ..., \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, ..., a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + ... + b_{51}$ and $s = a_1 + a_2 + ... + a_{51}$, then **[JEE (Adv)-2016 (Paper-2)]**

(A) $s > t$ and $a_{101} > b_{101}$

(B) $s > t$ and $a_{101} < b_{101}$

(C) $s < t$ and $a_{101} > b_{101}$

(D) $s < t$ and $a_{101} < b_{101}$

One or More Option(s) Correct Type Questions

4. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) **[JEE (Adv)-2013 (Paper-1)]**

(A) 1056

(B) 1088

(C) 1120

(D) 1332

5. If $I = \sum_{k=1}^{98} \int_k^{k+1} \dfrac{k+1}{x(x+1)} dx$, then **[JEE (Adv)-2017 (Paper-2)]**

(A) $I > \dfrac{49}{50}$

(B) $I < \dfrac{49}{50}$

(C) $I < \log_e 99$

(D) $I > \log_e 99$

6. Let a_1, a_2, a_3, … be an arithmetic progression with a_1 = 7 and common difference 8. Let T_1, T_2, T_3, … be such that T_1 = 3 and $T_{n+1} - T_n = a_n$ for $n \geq 1$. Then, which of the following is/are TRUE?

[JEE (Adv)-2022 (Paper-1)]

(A) $T_{20} = 1604$

(B) $\sum_{k=1}^{20} T_k = 10510$

(C) $T_{30} = 3454$

(D) $\sum_{k=1}^{30} T_k = 35610$

7. Let **[JEE (Adv)-2022 (Paper-2)]**

$$\alpha = \sum_{k=1}^{\infty} \sin^{2k}\left(\frac{\pi}{6}\right)$$

Let $g : [0, 1] \to \mathbb{R}$ be the function defined by

$g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$

Then, which of the following statements is/are TRUE?

(A) The minimum value of $g(x)$ is $2^{7/6}$

(B) The maximum value of $g(x)$ is $1 + 2^{1/3}$

(C) The function $g(x)$ attains its maximum at more than one point

(D) The function $g(x)$ attains its minimum at more than one point

Linked Comprehension Type Questions

Paragraph for Q.Nos. 8 to 10

Let V_r denote the sum of the first r terms of an arithmetic progression (A.P.) whose first term is r and the common difference is $(2r - 1)$. Let $T_r = V_{r+1} - V_r - 2$ and $Q_r = T_{r+1} - T_r$ for r = 1, 2,..... **[IIT-JEE-2007 (Paper-1)]**

Choose the correct answer :

8. The sum $V_1 + V_2 + ... + V_n$ is

(A) $\frac{1}{12}n(n+1)(3n^2 - n + 1)$

(B) $\frac{1}{12}n(n+1)(3n^2 + n + 2)$

(C) $\frac{1}{2}n(2n^2 - n + 1)$

(D) $\frac{1}{3}(2n^3 - 2n + 3)$

9. T_r is always

(A) An odd number

(B) An even number

(C) A prime number

(D) A composite number

10. Which one of the following is a correct statement?

(A) Q_1, Q_2, Q_3,.... are in A.P. with common difference 5

(B) Q_1, Q_2, Q_3,.... are in A.P. with common difference 6

(C) Q_1, Q_2, Q_3,.... are in A.P. with common difference 11

(D) $Q_1 = Q_2 = Q_3 = ...$

Paragraph for Q.Nos. 11 to 13

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

[IIT-JEE-2007 (Paper-2)]

Choose the correct answer :

11. Which one of the following statements is correct?

(A) $G_1 > G_2 > G_3 > ...$
(B) $G_1 < G_2 < G_3 < ...$
(C) $G_1 = G_2 = G_3 = ...$
(D) $G_1 < G_3 < G_5 < ...$ and $G_2 > G_4 > G_6 > ...$

12. Which one of the following statements is correct?

(A) $A_1 > A_2 > A_3 > ...$
(B) $A_1 < A_2 < A_3 < ...$
(C) $A_1 > A_3 > A_5 > ...$ and $A_2 < A_4 < A_6 < ...$
(D) $A_1 < A_3 < A_5 < ...$ and $A_2 > A_4 > A_6 > ...$

13. Which one of the following statements is correct?

(A) $H_1 > H_2 > H_3 > ...$
(B) $H_1 < H_2 < H_3 < ...$
(C) $H_1 > H_3 > H_5 > ...$ and $H_2 < H_4 < H_6 < ...$
(D) $H_1 < H_3 < H_5 < ...$ and $H_2 > H_4 > H_6 > ...$

Paragraph for Q.Nos. 14 and 15

Let $M = \{(x, y) \in R \times R : x^2 + y^2 \le r^2\}$, where $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}$, $n = 1, 2, 3, ...$ Let $S_0 = 0$ and, for $n \ge 1$, let S_n denote the sum of the first n terms of this progression. For $n \ge 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n, and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n.

[JEE (Adv)-2021 (Paper-2)]

14. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M. Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then

(A) $k + 2l = 22$
(B) $2k + l = 26$
(C) $2k + 3l = 34$
(D) $3k + 2l = 40$

15. Consider M with $r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is

(A) 198
(B) 199
(C) 200
(D) 201

Integer / Numerical Value Type Questions

16. Let S_k, $k = 1, 2, ..., 100$, denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100!} + \sum_{k=1}^{100} \left|(k^2 - 3k + 1)S_k\right|$ is **[IIT-JEE-2010 (Paper-1)]**

17. Let $a_1, a_2, a_3, ..., a_{11}$ be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, ..., 11$. If $\frac{a_1^2 + a_2^2 + ... + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + ... + a_{11}}{11}$ is equal to **[IIT-JEE-2010 (Paper-2)]**

18. Let $a_1, a_2, a_3, \ldots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^{p} a_i, 1 \le p \le 100$. For any integer n with $1 \le n \le 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n, then a_2 is **[IIT-JEE-2011 (Paper-1)]**

19. The minimum value of the sum of real numbers a^{-5}, a^{-4}, $3a^{-3}$, 1, a^8 and a^{10} with $a > 0$ is **[IIT-JEE-2011 (Paper-1)]**

20. A pack contains n cards numbered from 1 to n. Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k, then $k - 20 =$ **[JEE (Adv)-2013 (Paper-1)]**

21. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is **[JEE (Adv)-2014 (Paper-1)]**

22. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is **[JEE (Adv)-2015 (Paper-2)]**

23. The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side? **[JEE (Adv)-2017 (Paper-1)]**

24. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ..., and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, ... Then, the number of elements in the set $X \cup Y$ is _______. **[JEE (Adv)-2018 (Paper-1)]**

25. Let AP (a; d) denote the set of all the terms of an infinite arithmetic progression with first term a and common difference $d > 0$. If AP(1; 3) $\cap$ AP(2; 5) $\cap$ AP(3; 7) = AP(a; d), then $a + d$ equals ______.

[JEE (Adv)-2019 (Paper-1)]

26. Let m be the minimum possible value of $\log_3\left(3^{y_1} + 3^{y_2} + 3^{y_3}\right)$, where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^2)$ is ______

[JEE (Adv)-2020 (Paper-1)]

27. Let $a_1, a_2, a_3, \ldots$ be a sequence of positive integers in arithmetic progression with common difference 2. Also, let $b_1, b_2, b_3, \ldots$ be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c, for which the equality

$$2(a_1 + a_2 + \ldots + a_n) = b_1 + b_2 + \ldots + b_n$$

holds for some positive integer n, is ______ **[JEE (Adv)-2020 (Paper-1)]**

28. Let $l_1, l_2, \ldots, l_{100}$ be consecutive terms of an arithmetic progression with common difference d_1, and let $w_1, w_2, \ldots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2, where $d_1 d_2 = 10$. For each $i = 1, 2, \ldots, 100$, let R_i be a rectangle with length l_i, width w_i and area A_i. If A_{51} “ $A_{50} = 1000$, then the value of A_{100} “ A_{90} is ____________. **[JEE (Adv)-2022 (Paper-1)]**

29. The product of all positive real values of x satisfying the equation $x^{\left(16(\log_5 x)^3 - 68 \log_5 x\right)} = 5^{-16}$ is _____.

[JEE (Adv)-2022 (Paper-2)]

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